



Advanced Linear Algebra: Foundations to Frontiers

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1.3.6 Computing the matrix 1-norm and ∞ -norm



01.3.6 1-norm and inf-norm

fit width



The matrix 1-norm and matrix ∞ -norm are of great importance because, unlike the matrix 2-norm, they are easy and relatively cheap to compute.. The following exercises show how to practically compute the matrix 1-norm and ∞ -norm.

Homework 1.3.6.1. Let $A \in \mathbb{C}^{m \times n}$ and partition $A = \left(\begin{array}{c|c|c|c} a_0 & a_1 & \cdots & a_{n-1} \end{array} \right)$.

ALWAYS/SOMETIMES/NEVER: $\|A\|_1 = \max_{0 \leq j < n} \|a_j\|_1$.

▼ [Hint](#)

▼ [Answer](#)

▼ [Solution](#)

Let J be chosen so that $\max_{0 \leq j < n} \|a_j\|_1 = \|a_J\|_1$. Then

$$\begin{aligned}
& \|A\|_1 \\
&= \text{< definition >} \\
& \max_{\|x\|_1=1} \|Ax\|_1 \\
&= \text{< expose the columns of } A \text{ and elements of } x \text{ >} \\
& \max_{\|x\|_1=1} \left\| \begin{pmatrix} a_0 & a_1 & \cdots & a_{n-1} \end{pmatrix} \begin{pmatrix} \chi_0 \\ \chi_1 \\ \vdots \\ \chi_{n-1} \end{pmatrix} \right\|_1 \\
&= \text{< definition of matrix-vector multiplication >} \\
& \max_{\|x\|_1=1} \|\chi_0 a_0 + \chi_1 a_1 + \cdots + \chi_{n-1} a_{n-1}\|_1 \\
&\leq \text{< triangle inequality >} \\
& \max_{\|x\|_1=1} (\|\chi_0 a_0\|_1 + \|\chi_1 a_1\|_1 + \cdots + \|\chi_{n-1} a_{n-1}\|_1) \\
&= \text{< homogeneity >} \\
& \max_{\|x\|_1=1} (|\chi_0| \|a_0\|_1 + |\chi_1| \|a_1\|_1 + \cdots + |\chi_{n-1}| \|a_{n-1}\|_1) \\
&\leq \text{< choice of } a_J \text{ >} \\
& \max_{\|x\|_1=1} (|\chi_0| \|a_J\|_1 + |\chi_1| \|a_J\|_1 + \cdots + |\chi_{n-1}| \|a_J\|_1) \\
&= \text{< factor out } \|a_J\|_1 \text{ >} \\
& \max_{\|x\|_1=1} (|\chi_0| + |\chi_1| + \cdots + |\chi_{n-1}|) \|a_J\|_1 \\
&= \text{< algebra >} \\
& \|a_J\|_1.
\end{aligned}$$

Also,

$$\begin{aligned}
& \|a_J\|_1 \\
&= \text{< } e_J \text{ picks out column } J \text{ >} \\
& \|Ae_J\|_1 \\
&\leq \text{< } e_J \text{ is a specific choice of } x \text{ >} \\
& \max_{\|x\|_1=1} \|Ax\|_1.
\end{aligned}$$

Hence

$$\|a_J\|_1 \leq \max_{\|x\|_1=1} \|Ax\|_1 \leq \|a_J\|_1$$

which implies that

$$\max_{\|x\|_1=1} \|Ax\|_1 = \|a_J\|_1 = \max_{0 \leq j < n} \|a_j\|_1.$$

ALWAYS

Prove it for the real valued case first.

Homework 1.3.6.2. Let $A \in \mathbb{C}^{m \times n}$ and partition $A =$

$$\begin{pmatrix} \frac{\tilde{a}_0^T}{\tilde{a}_1^T} \\ \vdots \\ \tilde{a}_{m-1}^T \end{pmatrix}.$$

ALWAYS/SOMETIMES/NEVER:

$$\|A\|_\infty = \max_{0 \leq i < m} \|\tilde{a}_i\|_1 (= \max_{0 \leq i < m} (|\alpha_{i,0}| + |\alpha_{i,1}| + \cdots + |\alpha_{i,n-1}|))$$

Notice that in this exercise $\tilde{a}_i$ is really $(\tilde{a}_i^T)^T$ since $\tilde{a}_i^T$ is the label for the i th row of matrix A .

► [Hint](#) ► [Answer](#) ► [Solution](#)

Remark 1.3.6.1. The last homework provides a hint as to how to remember how to compute the matrix 1-norm and ∞ -norm: Since $\|x\|_1$ must result in the same value whether x is considered as a vector or as a matrix, we can remember that the matrix 1-norm equals the maximum of the 1-norms of the columns of the matrix: Similarly, considering $\|x\|_\infty$ as a vector norm or as matrix norm reminds us that the matrix ∞ -norm equals the maximum of the 1-norms of vectors that become the rows of the matrix.